Patent Summary: HEALTH-BAND Neuro

Title: Modular Neuromodulation and Gesture Control Wristband Ecosystem with Detachable Diagnostic Core and Pass-Through Charging **Inventor:** Srikanth Patchava **Affiliation:** Embedded Operating Systems Research Foundation (EoS) **Date:** May 24, 2026 **Status:** Ready for Provisional Filing

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Abstract of the Invention

The present invention relates to a modular wearable ecosystem comprising a flexible wristband apparatus (the “Strap”) and a detachable, rigid diagnostic core (the “Core”). The Strap features an integrated array of surface electromyography (sEMG) sensors for gesture recognition and neuromodulation electrodes for transcutaneous electrical nerve stimulation (TENS). The Core features a dual-connector pass-through architecture (e.g., USB-C male to USB-C female) allowing simultaneous data transmission and device charging. When docked, the Core provides processing, power, and environmental/breath diagnostic capabilities to the Strap. When detached, the Core functions as an autonomous inline diagnostic dongle. This modular architecture uniquely bridges episodic biochemical diagnostics with continuous neuro-muscular monitoring and therapy.

Background and Problem Solved

Current wearable devices are siloed into rigid categories: fitness trackers (passive optical monitoring), diagnostic dongles (episodic chemical/breath testing), and therapeutic bands (neuromodulation or gesture control). Integrating all these functions into a single wearable is mechanically prohibitive, as breathalyzers and USB connectors require rigid housings, while sEMG and neuromodulation require flexible, continuous skin contact.

The present invention solves this by introducing a **modular ecosystem**. The rigid, chemical/diagnostic components are housed in a detachable USB-C Core, while the flexible, neuro-muscular components are housed in the Strap. Furthermore, the Core introduces a novel “pass-through” charging architecture, allowing it to act as an inline environmental monitor when plugged between a mobile device and a power source.

Key Claims Summary

The patent application will include claims covering the following core innovations:

1. The Modular Electro-Mechanical Docking Architecture

A wearable apparatus comprising a flexible wristband containing a central docking receptacle, wherein the receptacle includes an internal data/power interface (e.g., female USB-C) configured to receive a rigid, detachable diagnostic core. The Core provides computational processing and power delivery to the active electrode arrays within the flexible wristband.

2. Pass-Through Charging and Inline Monitoring (The Core)

A diagnostic dongle featuring a dual-connector architecture comprising a male connector on a first end and a female connector on an opposing second end. The apparatus is configured to permit continuous power and data pass-through between a host device (e.g., a smartphone) and a power source, while simultaneously utilizing onboard environmental sensors (e.g., VOC, temperature) and acting as a mass storage device.

3. Integrated sEMG and Neuromodulation Array (The Strap)

A flexible wristband apparatus comprising a circumferential array of electrodes capable of dual-mode operation:

- **Sensing Mode (sEMG):** Detecting motor unit action potentials for gesture recognition and human-computer interaction.
- **Therapeutic Mode (Neuromodulation):** Delivering targeted electrical pulses (TENS/tDCS) for pain management, tremor suppression, or autonomic nervous system regulation.

4. Unified Breath and Wrist Interface

A method of operation wherein the user can perform a breath analysis test (e.g., Blood Alcohol Content or VOC detection) by exhaling into a recessed Venturi channel located on the superior surface of the docked Core, without removing the apparatus from the wrist.

5. Shared Power and Processing Topology

An architecture wherein the flexible wristband contains no primary battery or primary microcontroller, relying entirely on the docked Core for power delivery and signal processing, thereby reducing the manufacturing cost and weight of the wearable strap.

Strategic Value of this Patent

Filing this patent creates a highly defensible intellectual property perimeter around the entire HEALTH-KEY brand.

1. **Patent 1 (App. 64/073,334):** Protects the standalone USB-C diagnostic dongle (HEALTH-KEY ULTRA).
2. **Patent 2 (This Document):** Protects the modular wristband ecosystem and the pass-through charging architecture (HEALTH-BAND Neuro).

This dual-patent strategy ensures that competitors cannot copy the standalone dongle, nor can they copy the method of docking a diagnostic dongle into a gesture-control wristband.